

CHAPTER 4

ENDOCRINE RESPONSES TO RESISTANCE EXERCISE AND TRAINING

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After completing this chapter, you will be able to

- identify basic concepts of endocrinology and explain the endocrine system's role in human physiology;
- describe the major hormones, including their sources, target tissues, and physiological effects;
- analyze and explain the physiological roles of anabolic and catabolic hormones in the body's metabolism and response to stress;
- examine and describe how resistance exercise and training elicit specific hormonal responses, differentiating between various types of resistance exercises;
- evaluate the impact of different resistance exercise workouts on hormonal responses, using scientific evidence to support an analysis; and
- use an understanding of human endocrine responses to develop effective workouts and training programs that optimize

hormonal benefits for health and performance.

The endocrine system interacts with the nervous, immune, and musculoskeletal systems to create a network vital for health and athletic performance. Hormones regulate cellular activities, including muscle synthesis, recovery, and overall physiological well-being, by balancing anabolic and catabolic processes. Strength and conditioning professionals must understand the endocrine system to design training programs that enhance performance, recovery, and injury prevention. This knowledge allows for manipulating hormonal responses to exercise, supporting adaptations, and preventing overtraining. The system's interplay has far-reaching effects on athletic performance and broader health concerns such as metabolic syndrome, insulin resistance, and cardiovascular health.

Historical Perspective on Interest in the Endocrine System

The significance of the endocrine system in strength and conditioning gained prominence with the rise of anabolic drug use in the 1940s and 1950s (7, 58). Fahey and colleagues were the first to investigate how resistance exercises affected endogenous hormone concentrations, specifically testosterone, across different sexes, ages, and training statuses, paving the way for ongoing research into hormonal responses to resistance exercise and training (23). In 1988, a series of reviews were published from the first symposium on the physiology of resistance exercise and training at the American College of Sports Medicine's annual meeting. As part of that symposium set of papers, Kraemer (97) overviewed the sparse number of studies that had been done on the various hormonal responses to resistance exercise and training. This stimulated a line of hormonal research that continues today to explore further the mechanisms behind these acute responses and training adaptations (46). Now the body of literature consists of thousands of studies on this topic.

Endocrine Terminology, Functions, and Mechanisms

Grasping the effects of resistance exercise on hormonal responses requires a solid understanding of endocrine system terminology. While the learning curve is significant, becoming familiar with this language is vital for effectively translating scientific insights into practical training strategies. **Hormones**, the chemical messengers or signal molecules, are synthesized, often stored, and released into the blood by **endocrine glands** and certain other cells ([figure 4.1](#), [table 4.1](#)). Neurons also synthesize, store, and secrete neurotransmitters, which may perform hormonal functions, a study known as **neuroendocrinology**—exploring interactions between the nervous and endocrine systems. Typically, endocrine glands are prompted to release hormones by a chemical signal received by receptors on the gland or by direct neural stimulation. For example, the adrenal medulla, the inner part of the adrenal gland, releases the hormone epinephrine in response to sympathetic neural stimulation from the brain (68, 74, 87, 129). Meanwhile, the outer layer of the adrenal cortex produces and releases cortisol when stimulated by adrenocorticotrophic hormone (ACTH) from the pituitary gland (75, 78, 90). Once released, these hormones travel in the bloodstream to specific receptors located on the surface (for peptide hormones) or inside the **target tissue cells** (for **steroid** and **thyroid hormones**) (4, 5, 6, 8, 22, 39), facilitating a targeted and efficient response.

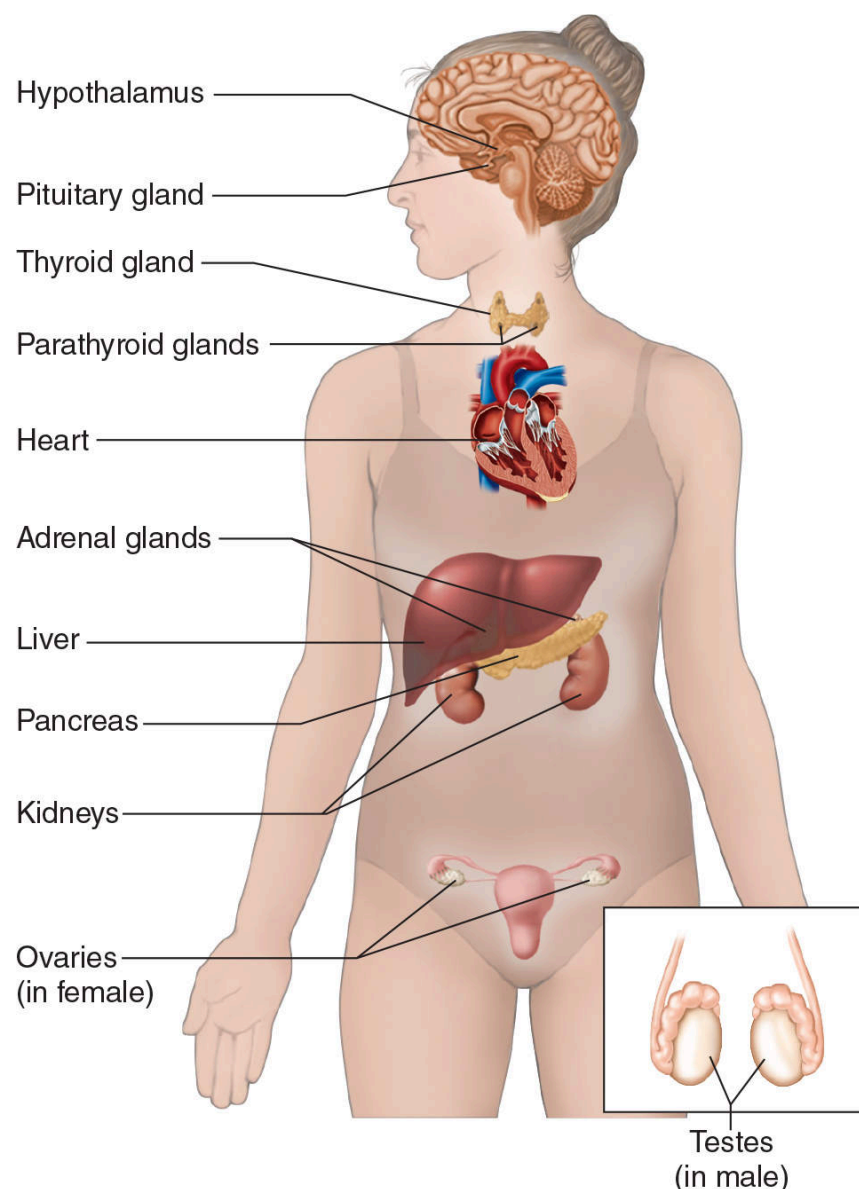


FIGURE 4.1 The principal endocrine glands of the body, along with other glands that secrete hormones.

TABLE 4.1 Endocrine Glands and Selected Hormones

Endocrine gland	Hormone	Selected physiological actions
Anterior pituitary gland	Growth hormone(s)	Stimulates insulin-like growth factor-1 secretion from the liver, protein synthesis, growth, and metabolism; other aggregates of the growth hormone (GH) also have biological function and make up the more complex super family of GH

Endocrine gland	Hormone	Selected physiological actions
	Adrenocorticotrophic hormone	Stimulates glucocorticoid secretion from the adrenal cortex
	Beta-endorphin	Stimulates analgesia
	Thyroid-stimulating hormone	Stimulates thyroid hormone secretion from the thyroid gland
	Follicle-stimulating hormone	Stimulates growth of follicles in ovary and seminiferous tubules in testes; stimulates ovum and sperm production
	Luteinizing hormone	Stimulates ovulation as well as secretion of sex hormones in the gonads (ovaries and testes)
	Prolactin	Stimulates milk production in mammary glands; maintains corpora lutea and secretion of progesterone
Posterior pituitary gland	Antidiuretic hormone	Increases contraction of smooth muscle and reabsorption of water by kidneys
	Oxytocin	Stimulates uterine contractions and release of milk by mammary glands
Thyroid gland	Thyroxine	Stimulates oxidative metabolism in mitochondria and cell growth
	Calcitonin	Reduces calcium phosphate levels in blood
Parathyroid glands	Parathyroid hormone	Increases blood calcium; decreases blood phosphate; stimulates bone formation
Pancreas	Insulin	Reduces blood glucose concentrations via promotion of glucose uptake by cells; promotes glycogen storage; suppresses fat oxidation and gluconeogenesis; is involved in protein synthesis
	Glucagon	Increases blood glucose levels
Adrenal cortex	Glucocorticoids (cortisol, cortisone, and so on)	Catabolic and anti-anabolic: promote protein breakdown and inhibit amino acid incorporation into proteins; conserve blood glucose concentrations via stimulation of conversion of proteins into carbohydrates (gluconeogenesis);

Endocrine gland	Hormone	Selected physiological actions
		suppress immune cell function; promote fat oxidation
	Mineralocorticoids (aldosterone, deoxycorticosterone, and so on)	Increase body fluids via sodium–potassium retention
Liver	Insulin-like growth factors	Increase protein synthesis in cells
Adrenal medulla	Epinephrine	Increases cardiac output; increases blood sugar and glycogen breakdown and fat metabolism
	Norepinephrine	Has properties of epinephrine; also constricts blood vessels
	Proenkephalin fragments (e.g., peptide F)	Enhance immune cell function, have analgesia effects
Ovaries	Estradiol	Stimulates development of female sex characteristics
	Progesterone	Stimulates development of female sex characteristics and mammary glands; maintains pregnancy
Testes	Testosterone	Anabolic and anticatabolic: promotes amino acid incorporation into proteins and inhibits protein breakdown; stimulates growth and development and maintenance of male sex characteristics
Heart (atrium)	Atrial peptide	Regulates sodium, potassium, and fluid volume
Kidney	Renin	Regulates kidney function, permeability, solute

Understanding Signaling Concepts

Different types of workouts trigger specific, complex cellular endocrine signaling processes, thus highlighting the importance of the selection for each of the five acute program variables and then the total amount of work performed in a workout (70) for the endocrine response. Deoxyribonucleic acid (DNA) primarily directs protein synthesis, but the regulation and expression of DNA affect many other cellular functions and the synthesis of various molecules essential for life. Hormonal signaling processes begin when glands release hormones during, or in anticipation of, a resistance exercise workout. These hormones then bind to specific receptors in muscle cells and other cells. This binding initiates a cascade of responses, generally resulting in altered cellular metabolism or DNA transcription modulation, which can affect protein synthesis. It is essential to know that every cell and many subcellular organelles have genomic and non-genomic receptors, and this is where the hormone signals its message to make proteins. Here is a simplified breakdown of the signaling process for skeletal muscle:

1. *Stimulating the muscle.* Resistance exercise activates muscles, prompting a hormonal response.
2. *Hormonal response.* Hormones are released and travel to muscle cells.
3. *Hormone–receptor interaction.* Hormones bind to specific receptors in the cell membrane of muscle cells (peptide hormones) or inside the muscle cells (steroid hormones), signaling the need for action inside the cell.
4. *Intracellular signaling.* This binding triggers a series of events that lead to DNA signaling inside the nucleus.
5. *DNA translation.* DNA instructions are transcribed to messenger ribonucleic acid (mRNA), which carries the instructions to the cell's protein-making machinery (ribosome).

6. *Protein production.* The mRNA instructions are translated into specific proteins, the building blocks for muscle repair and growth.
7. *Muscle growth and repair.* Regular training uses this muscle development and repair mechanism, which is essential for developing lean muscle mass and athletic performance.

Functions of the Endocrine System in Resistance Exercise and Training

The major functions of the endocrine system can be classified into the following key hormonal actions:

- *Anabolic responses (to build up)*. Hormones such as testosterone, insulin-like growth factor-1 (IGF-1), and growth hormone play critical roles in muscle growth and repair.
- *Permissive actions (to allow)*. Hormones such as insulin and thyroid hormones enable critical metabolic and physiological activities by enhancing nutrient uptake and promoting essential chemical reactions in muscles, thereby supporting muscle growth and overall metabolic functions.
- *Catabolic processes (to break down)*. For example, cortisol helps manage energy during stress but can hinder muscle growth if not properly managed through adequate rest and recovery. This emphasizes the balance needed in training regimens.

Types of Hormonal Release Mechanisms

Hormones are typically released into the bloodstream via the classic endocrine pathway after gland stimulation. Hormones can also operate through intracrine, autocrine, and paracrine mechanisms, which do not rely on the bloodstream. Intracrine and autocrine actions involve hormones either binding to internal receptors within their producing cell or being released to bind to surface receptors on the same cell. An example of this is insulin-like growth factor-1 (IGF-1) and its splice variant mechano growth factor (MGF), which muscle fibers produce in response to mechanical stress or interaction with growth hormones, affecting the muscle cell directly. Paracrine action occurs when hormones affect adjacent cells, again bypassing the bloodstream.

Binding Proteins

Hormones are often transported in the blood by various binding proteins (4, 6), which serve as a temporary storage location and protect the hormones from degradation, thereby extending their lifetime in the body (called the **half-life**). It is important to note that hormones typically exert their effects only when they are released, and thus are “free,” from these proteins. However, some binding proteins, such as sex hormone–binding globulin (SHBG), which binds testosterone and estrogen, can initiate cellular responses themselves by activating pathways, including the cyclic adenosine monophosphate (cAMP) pathway (29). These interactions are crucial for understanding endocrine regulation and continue as an area of study (101).

Target Cells and Tissues

Hormones influence a wide array of cells by interacting with receptors throughout the body (1, 2, 57, 59, 62, 63). While our focus is on skeletal muscle tissue, it is important to recognize that resistance exercise–induced hormone release also affects bone, other connective tissues, the kidneys, the liver, immune cells, and neural tissues, including the brain. Interestingly, not only do these target cells in the different tissues adapt to the exercise stress, but their receptors also gain a positive adaptive effect with training.

Importance of Recruitment of Muscle Tissue

Henneman's size principle shows that as the resistance increases (i.e., heavier loads), more and larger muscle fibers are recruited to meet the increased force demand. This principle outlines the recruitment order of motor units—from smaller, slow-twitch fiber motor units to larger, fast-twitch fiber motor units. The heavier the resistance, the greater the number and size of motor units recruited in a muscle to produce force (77). While anabolic and catabolic hormones maintain homeostatic balance by interacting with all cells, stress causes hormonal changes that reflect the body's need to meet new metabolic demands and recovery needs. However, only recruited motor units adapt to the training stimulus and are affected by the endocrine system's increase in anabolic hormones, likely due to the increased activation of more receptors. Interestingly, and of importance, increased catabolic hormone responses to stress or distress can affect all tissues even if not recruited for exercise performance. A catabolic state is highly detrimental for an athlete, especially when cortisol levels remain chronically elevated. Combined with the presence of reactive oxygen species (free radicals) generated by the body's inflammatory response, this condition significantly impairs recovery and hampers performance (77).

Muscle as the Target for Hormone Interactions

Resistance exercise and training leads to muscle remodeling through a complex network of metabolic and cellular signals. This process involves muscle fiber damage, an inflammatory response, and the breakdown of damaged proteins, ultimately synthesizing new proteins integrated into muscle structures. The inflammatory phase, which involves the immune system, including T and B cells, is significantly influenced by hormonal activity, emphasizing the role of **neuroendocrine immunology** in understanding how the nervous, endocrine, and immune systems interact in muscle adaptation (77).

Hormones regulate the balance between protein synthesis and degradation, playing a critical role in muscle adaptation after resistance exercise. They stimulate the production of contractile proteins like actin and myosin, as well as structural proteins such as titin, which are essential for muscle function. **Anabolic hormones**, including insulin, insulin-like growth factors (IGFs), testosterone, and GH, are particularly vital because they drive tissue growth and counteract the effects of **catabolic hormones** like cortisol and progesterone, which can lead to protein degradation. Furthermore, hormones like thyroid hormones play a supportive, permissive role, facilitating the actions of other hormones. The potentially negative effects of cortisol in recovery, such as inhibiting immune cell activity and disrupting pathways like the Akt/mechanistic target of rapamycin (mTOR) pathway—important for initiating mRNA translation—are also notable, demonstrating the complex interactions between hormones and muscle fibers that influence muscle function and adaptation to training.

Role of Receptors in Mediating Hormonal Changes

The effectiveness of hormonal signals depends entirely on the presence of hormone-specific receptors on or in the target cells. This ensures that each hormone only affects its intended cells and tissues (those that express the specific receptors). Such specificity is crucial for maintaining precise hormonal regulation within the body. Since many hormones cannot directly cross the cell membrane, they rely on binding to membrane receptors that cause **secondary messengers** to transmit signals internally, including reaching the DNA in the cell's nuclei. These receptors, located either in the cell membrane for **polypeptide hormones** or within the cytosol for steroid and thyroid hormones, allow various cells, from muscle fibers to immune and brain cells, to respond to hormonal signals.

A fundamental principle in endocrinology, often described through the **lock-and-key model** (24), explains how a hormone (the key) must precisely match a receptor (the lock) to trigger a response ([figure 4.2](#)). However, this model oversimplifies the complexity of hormone–receptor interactions. For example, **cross-reactivity** allows receptors to engage with hormones other than their primary matches, and **allosteric binding** can modify how a receptor responds to its primary hormone (such as blocking binding to the receptor and thus signaling). Additionally, receptors might have allosteric sites where nonhormonal substances can enhance or reduce the hormonal response. Some hormones, like GH, might even need to aggregate into specific forms (multiple hormones bound together in a “cluster”) to effectively signal through their receptors, as shown in studies using the tibial line assay (40), indicating that specific molecular configurations are essential for their activity.

Adaptation Limits and Receptor Sensitivity

When a muscle fiber has reached its capacity for protein incorporation, or when a chronically elevated concentration of a hormone causes overstimulation, receptors may become desensitized or unresponsive—a condition known as receptor **downregulation**; a classic example of this phenomenon is insulin resistance, which can be caused by chronically elevated circulating insulin concentrations. The reduced signal from the hormone following overstimulation signifies a reduced ability of the hormone to interact with its receptor or downregulation of the number of receptors, which can effectively stop further cellular responses and adaptations (37, 122). As such, receptors can adjust their affinity for binding, and the number of available receptors can vary. These changes affect cellular adaptation as significantly, if not more, as changes in hormone concentrations. For example, heavy resistance overtraining can lead to a downregulation of beta-2 receptors, affecting their interaction with epinephrine and reducing muscle contractile force (37). Exploring how receptor characteristics alter in response to training and affect muscle adaptation is key to understanding the complex interactions between hormones and exercise.

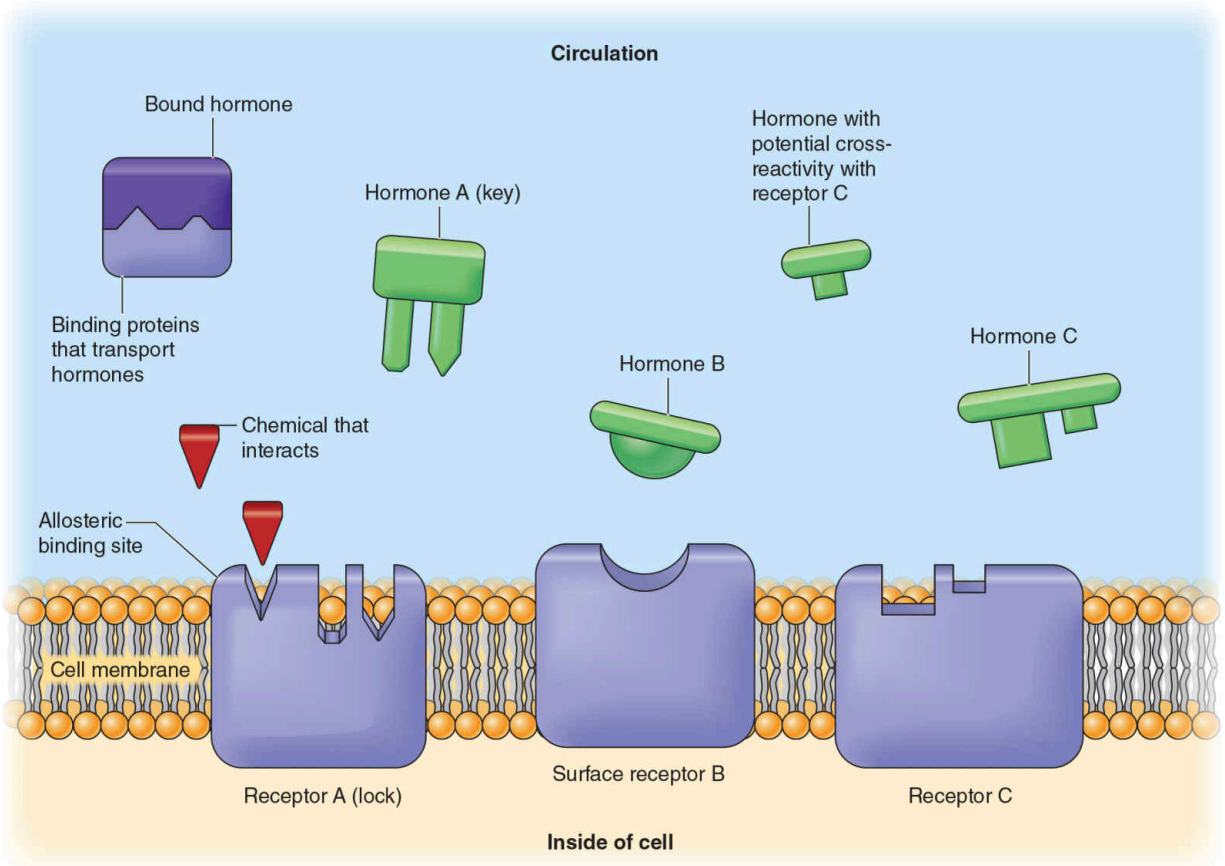


FIGURE 4.2 A schematic representation of the classic lock-and-key theory for hormonal action at the cell receptor level.

Categories of Hormones

Hormones can be classified into three main categories based on their molecular structure: steroid, peptide (or polypeptide), and amine hormones.

Steroid Hormone Interactions

Steroid hormones, including cortisol, testosterone, and estradiol produced by the adrenal cortex and/or gonads, can diffuse across cell membranes due to their lipophilic (fat-soluble) nature. They engage with specific receptors inside the cell to form a **hormone–receptor complex (H-RC)**. This complex undergoes a conformational change, activates, and pairs with another H-RC. It then moves to the nucleus, interacts with the DNA, and promotes the transcription process for specific proteins by “unzipping” DNA and identifying regulatory elements of the targeted gene (18, 117). The resulting mRNA is then translated by ribosomes in the sarcoplasm into the intended proteins. The steroid hormone–induced protein synthesis process is depicted in [figure 4.3](#).

Testosterone’s influence on muscle cells encompasses more than its genomic effects. In addition to the primary nuclear receptor, a membrane receptor for testosterone has also been described in detail (128). Such a membrane receptor allows testosterone to affect the cell through rapid and nongenomic interactions (i.e., those that do not involve DNA transcription in the nuclei) within various other cellular compartments. These interactions play a crucial role in muscle adaptation, recovery, and growth by facilitating the production of essential contractile and structural proteins, modulating calcium flux in muscle cells, altering membrane fluidity, and activating key signaling pathways (e.g., the MAPK/ERK pathway). Such nongenomic actions underscore testosterone’s multifaceted role in muscle development. Understanding testosterone’s genomic and nongenomic effects is vital for a

comprehensive grasp of steroid hormones' impact on muscle biology (91).

Polypeptide Hormone Interactions

Polypeptide hormones are composed of chains of amino acids and include hormones such as GH and insulin. Due to their non-lipophilic nature, polypeptide hormones cannot cross the cell membrane, and, therefore, their effects are mediated by secondary messengers within the cell. When these hormones bind to their membrane receptor, they cause a conformational change that allows the receptor to transmit the hormonal signal inside the cell. This transmission initiates a cascade of signaling events that, among others, can affect metabolic activities, influence DNA transcription, or kick-start mRNA translation at the ribosome. For example, insulin activates a pathway that results in glucose transporters (specifically GLUT4) migrating from the cytosol to the cell membrane, enhancing the cell's uptake of glucose (45). This is crucial for managing blood glucose levels and cellular substrate availability for ATP production.

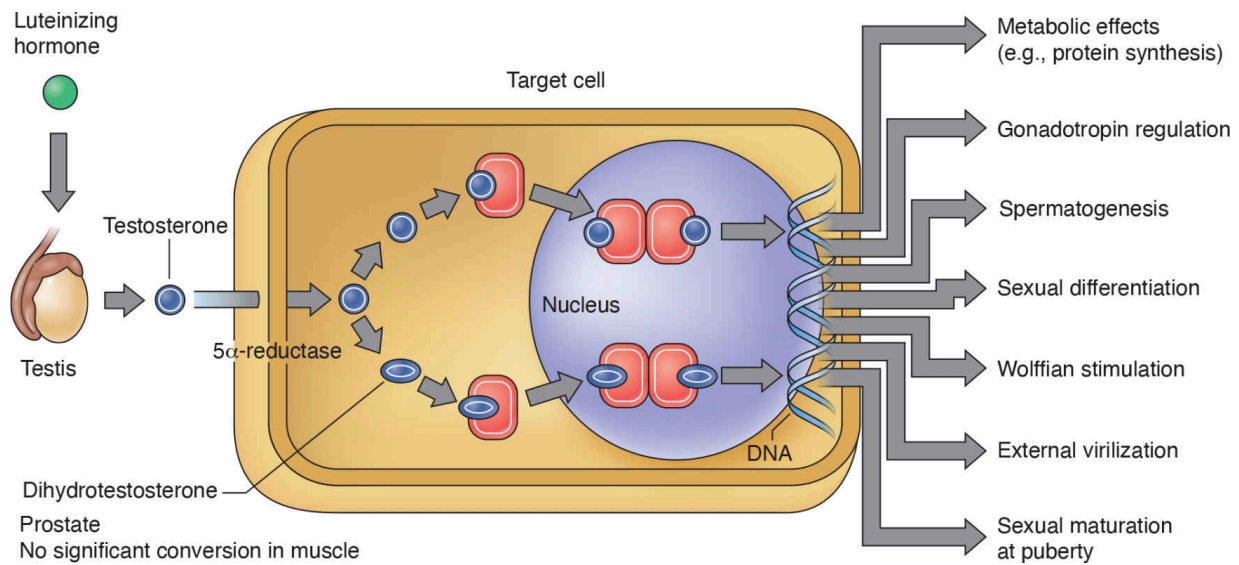


FIGURE 4.3 A simplified model of steroid hormone action. The figure illustrates how steroid hormones like testosterone in skeletal muscle or dihydrotestosterone in sex-linked tissues migrate into target cells. Although the pathways for both hormones are represented, each targets only one specific cell type but they are shown together for clarity. Each hormone pathway leads to different physiological outcomes, emphasizing the selective nature of steroid–hormone interactions.

[Figure 4.4](#) depicts how a typical polypeptide hormone might interact with the cell nucleus through the cytokine-activated Janus kinase (JAK)/signal transducer and activator of the transcription (STAT) signaling pathway. The JAK/STAT pathway transmits signals from various polypeptide hormones into the cell. However, like testosterone, polypeptide hormones may also have nongenomic actions. For instance, GH stimulates lipolysis, thus sparing muscle glycogen during and after a workout (110).

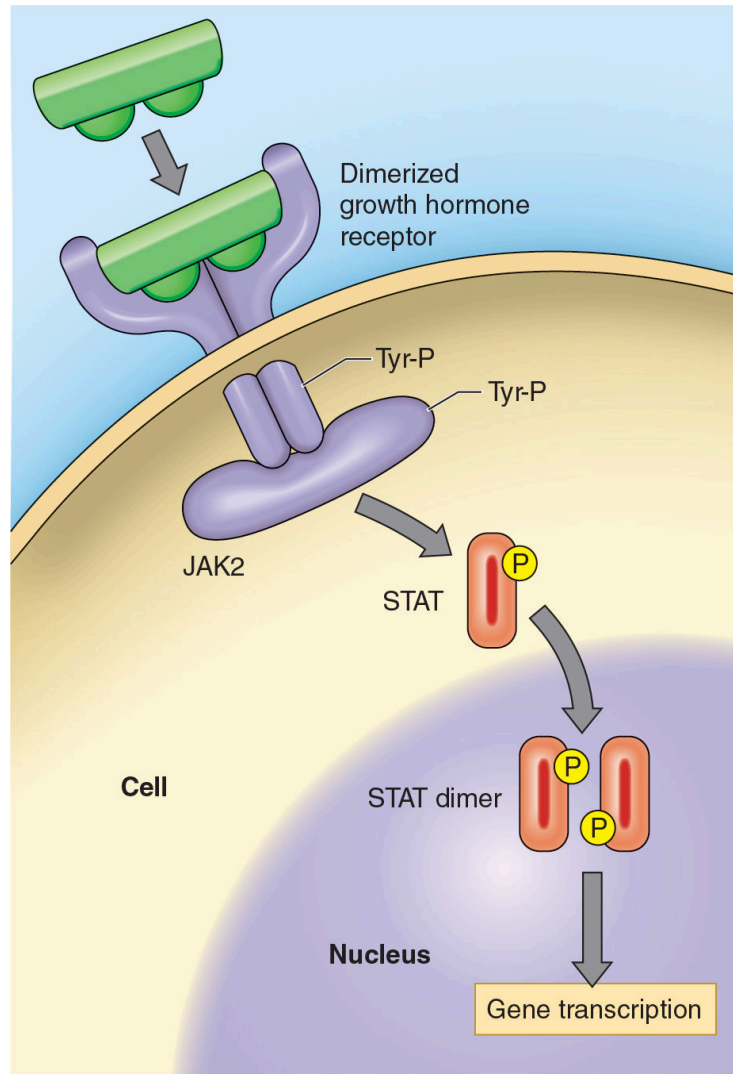


FIGURE 4.4 Polypeptide hormone interactions with receptors. Tyr-P = tyrosinase-related protein.

Amine Hormone Interactions

Amine hormones are indeed synthesized from amino acids, including those derived from tyrosine (such as epinephrine, norepinephrine, and dopamine) and tryptophan (such as serotonin), and primarily exert their effects by binding to membrane receptors. These hormones typically act through secondary messenger systems, similar to polypeptide hormones. For example, epinephrine and norepinephrine interact with G protein-coupled receptors to initiate signaling cascades involving cAMP or other secondary messengers.

Training-Mediated Hormonal Responses and Mechanisms

Electrical, chemical, and hormonal signals are transmitted from the brain, active muscles, and other tissues to various endocrine glands, preparing the body's physiological systems for the demands of the workout (20, 23, 28, 32, 35, 49, 75, 80, 81). Heavy resistance training over months to years significantly enhances musculature size, strength, and power, mediated partly through adaptive hormonal responses (47, 50, 70, 73, 76, 88, 95). These adaptations arise from increased levels of anabolic hormones such as testosterone, growth hormone, and insulin-like growth factor-1 (IGF-1) that acutely rise during exercise and remain elevated after exercise. Additionally, resistance exercise leads to an upregulation of hormone receptors in muscle tissues, which increases the muscle's sensitivity to these hormones and enhances protein synthesis and muscle hypertrophy. The structural changes in muscle and neural adaptations contribute to increased strength and power, including improvements in the rate of force production and alterations in muscle fiber recruitment patterns. Resistance training also positively affects bone density and the strength of connective tissues, supporting increased muscular strength and overall physical performance. One critical pathway in muscle protein synthesis activated by resistance training is the mammalian target of the rapamycin (mTOR) pathway, which is crucial for muscle growth. These complex interactions result in adaptive changes in skeletal muscle and other tissues, enhancing the activation of muscle functions over time.

Acute hormonal releases reflect the nature and intensity of physiological stress, such as insulin-related metabolic demands, prompting adjustments in resting metabolism. These coordinated hormonal responses are critical for adapting to training programs, influencing tissue remodeling, and promoting anabolic actions.

However, excessive exercise stress may lead to catabolic dominance due to factors like impaired hormone receptor binding or receptor downregulation in muscle tissue, highlighting the importance of appropriate exercise stimulus for beneficial hormonal responses (18, 25, 26, 27, 30, 69, 96, 99, 108).

Interpreting Hormonal Changes in Peripheral Blood

Understanding exercise effects on the body goes beyond physical observations; it includes reading studies that monitor biochemical changes such as hormone fluctuations in blood, muscle, and surrounding fluids (114). These changes provide insights into the body's stress response and adaptation to exercise. However, interpreting these changes can be complex, because hormone levels in the blood are just one aspect of a broader hormonal response that also includes neural and nutritional factors such as being fasted or not and if proteins and carbohydrates are ingested around a workout. Increases in blood hormone levels generally indicate enhanced release from endocrine glands, potentially boosting hormone–receptor interactions in target tissues. Yet the actual impact depends on the state and quantity of the receptors—whether they are primed to bind hormones and initiate crucial processes involved in protein synthesis, including activating pathways such as mTOR (11, 13, 14, 66). Conversely, decreases in hormone levels might suggest increased tissue uptake, faster hormone breakdown, or reduced secretion.

Several physiological factors, including circadian rhythms, fluid volume changes, tissue clearance rates, venous pooling, and interactions with binding proteins, also influence observed circulating hormone concentrations after exercise. These factors determine the hormone's efficacy in interacting with receptors and triggering cellular responses, underlining the complexity of hormone function and its role in exercise adaptations.

While increased hormone levels after exercise indicate heightened glandular activity, they do not directly correlate with muscle gains. Muscle size and strength can improve without significant changes in hormone levels by, for example, using heavy 1RM to 3RM zones with long rest periods of over 3 minutes. Such resistance loads recruit

maximal amounts of motor units and enhance receptor binding. However, such workouts' relatively low metabolic demands typically result in only modest, if any, increases in blood concentrations of testosterone, IGF-1, and growth hormone. Nevertheless, slight hormonal changes, combined with paracrine and autocrine signaling, are generally sufficient to effectively signal via receptors and other molecular pathways within tissues in the neuromuscular system to stimulate growth and repair. Conversely, a workout that involves 8RM to 10RM resistances, short rest periods of 1 to 2 minutes, and a higher volume of work produces greater metabolic demands on the body. The metabolic stress is evidenced by significant increases in blood lactate and a surge in both anabolic and catabolic hormones. These hormonal changes signal skeletal muscle receptors effectively, even with the recruitment of fewer motor units using lighter resistances. Periodization strategies that incorporate a range of different resistances over time ensure effective muscle activation across various workout styles, where hormones help mediate changes. Such workout design variations reflect the complex nature of exercise-induced hormonal activation and its contribution to adaptive changes in the body.

Hormone responses in the blood are tightly linked to the characteristics of the resistance exercise protocol.

Adaptations in the Endocrine System From Resistance Training

Resistance training affects muscles and connective tissues and significantly affects the other physiological systems, including the endocrine system. This type of training also trains the endocrine glands, with the gland's involvement dependent on the demands of the motor units activated during exercise. The endocrine system's adaptations to resistance training include changes in the following:

- Hormone synthesis and storage
- Hormone transport via binding proteins
- Clearance and degradation rates of hormones
- Fluid shifts from blood to tissues during exercise
- Hormone–receptor binding affinity
- The number of tissue receptors
- Glandular cell changes
- The intensity of signals sent to cell nuclei by hormone–receptor complexes
- Cellular interactions influencing muscle protein production

Hormones are secreted to maintain homeostasis, regulating physiological functions during both the acute stress of a workout and chronic stress from long-term training, which activates these homeostatic mechanisms (17, 33, 34, 41, 49, 56, 125, 126, 130). These adaptations might lead to subtle changes in resting hormone levels, demonstrating the dynamic responsiveness of the endocrine system to physical training.

Primary Anabolic Hormones in Muscle Development

Key anabolic hormones that drive muscle tissue growth and remodeling include testosterone, growth hormone, and insulin-like growth factors (IGFs), with additional roles played by insulin and thyroid hormones (25, 26, 27, 28, 38, 91).

Testosterone's Role in Muscle Adaptation

Testosterone, the primary androgen affecting skeletal muscle, and dihydrotestosterone, which acts mainly on sex-linked tissues, play crucial roles in the body. While circulating testosterone levels indicate anabolic activity, the actual binding of testosterone to its receptor is vital for stimulating muscle growth. Binding to androgen genomic and non-genomic receptors activates various signaling pathways that enhance gene expression for muscle hypertrophy and suppress those involved in protein degradation. Therefore, increased testosterone levels due to increased motor unit activation indicate metabolic demands beyond homeostatic conditions and are associated with enhanced receptor binding (124, 133).

Testosterone also indirectly affects muscle growth by promoting the release of growth hormone from the pituitary, which influences muscle protein synthesis. This hormone's relationship with growth hormone is either permissive or synergistic in promoting protein synthesis (103). Furthermore, testosterone affects the nervous system by interacting with neuron receptors, altering neurotransmitter amounts, and influencing structural protein changes within muscle cells (9, 67, 91).

After secretion from the primary production sites, the testes in men and the ovaries and adrenal cortex in women, testosterone is carried to target tissues by binding proteins, primarily sex hormone-binding globulin and albumin. At the target site, testosterone dissociates from these proteins and crosses the cell membrane to

bind to intracellular androgen receptors, directly influencing DNA transcription. Testosterone also binds to membrane receptors, enabling rapid intracellular effects like calcium release (21, 131).

Understanding the varied impacts of testosterone and its interaction with exercise conditions reveals how differential forces on cell membranes during resistance training or feedback mechanisms to the brain can alter cellular actions and receptor interactions (18). Despite increases in testosterone concentrations during high-intensity aerobic endurance exercise, significant hypertrophy does not typically occur. The lack of hypertrophy may be due to oxidative stress, which can lead to reduced type I muscle fiber size to optimize oxygen transport distances to mitochondria. High-intensity endurance training, including intervals performed alone, has been shown to lead to decreases in type I muscle fibers (88).

Influence of Sex Hormones

Testosterone, while primarily recognized as a male sex hormone, also plays a significant role in women, who naturally have only about 5% to 10% of the circulating testosterone levels found in men due to the absence of testes. Research on testosterone responses in women following resistance exercise has yielded mixed results. Most studies have not shown a notable acute increase in total testosterone after workouts; if increases are observed, they are generally small (15, 23, 51, 56, 81, 134). However, some studies note these increases specifically in free testosterone, which is not bound to binding proteins and thus can interact freely with cells (133).

A study found that younger women experienced a modest but significant increase in serum testosterone after performing six sets of squats at a 10RM with 2 minutes of rest between sets at 80% of the 1 RM (111). Additionally, Vingren and colleagues observed that highly resistance-trained men and women exhibited acute rises in free testosterone following heavy resistance exercises. However, the values in women were significantly lower than those in men (133).

While women have lower baseline testosterone levels, exercise can still influence their hormone levels.

Moreover, individual variations exist among women, with some producing higher concentrations of adrenal androgens. A study comparing active women with inactive controls found differences in baseline testosterone levels, indicating that regular exercise might influence these levels (15). Despite these findings, other research has not consistently demonstrated changes in serum testosterone concentrations in women with training (23, 51, 56, 81, 134), highlighting the complexity of hormonal responses to resistance exercise in women.

While testosterone levels in women are lower, the dynamic interaction between testosterone, SHBG, DHT, and androgen receptors facilitates significant responses to resistance exercise. These interactions can include muscle mass and strength increases, essential for physical health and metabolic function. The response of androgen receptors to even small increases in testosterone and the compensatory role of other androgens like DHT emphasize the complex but critical nature of androgens in female physiology and resistance exercise.

An interplay between free and total testosterone is critical for optimizing exercise outcomes. Free testosterone, which represents only 0.5% to 2% of total testosterone, is vital for direct interaction with muscle tissues. Younger men generally exhibit higher postexercise increases in free testosterone compared to older men, indicating a greater possibility for muscle tissue interaction (82). According to the free hormone hypothesis, the free, unbound hormone interacts with target tissues. However, bound hormones may also enhance hormone delivery to these tissues, potentially benefiting younger individuals more significantly (19). Due to women's relatively low total testosterone levels, changes in hormone-binding proteins could substantially regulate hormone activity, suggesting that these binding proteins might act as bioactive hormones (120).

Exercise-Induced Responses

Men's testosterone levels are generally highest in the morning and gradually decrease throughout the day. However, this does not mean that morning workouts are more effective than afternoon workouts. The effectiveness of exercise depends more on the availability of hormone receptors in both men and women, which is not directly linked to the body's circadian rhythm. Exercise can trigger increases any time, with the overall response related to intensity, volume, and rest periods between sets (17, 23, 44, 48, 81, 134). However, the magnitude of the testosterone's response is, as with all hormones, dependent on where blood is sampled in the hormone's circadian cycle (e.g., testosterone is higher in the morning than in the late afternoon), and thus, it is important to understand where a measure of a hormone was made and if the study has a resting control for the time of day to determine the magnitude above resting levels accurately. Its circadian cycle is important for the interpretation of the results of a study. Most studies have used the morning time frames.

Testosterone dynamics in young males, particularly those under 18, are influenced significantly by puberty, affecting their responses to exercise. Prepubescent boys produce minimal testosterone, which limits exercise-induced hypertrophy. Adolescent males can enhance serum testosterone levels through exercises that engage large muscle groups like squats and deadlifts, using heavy resistance of 85% to 95% of their 1RM, and maintaining high exercise volume with short rest intervals between sets, but the levels in women were dramatically lower ([figure 4.5](#); 133). It took two years of training in young weightlifters (14-17 years) before an exercise-induced testosterone effect was observed. While resistance training may not raise resting testosterone levels, repeated exposure to resistance training could help younger boys develop their hypothalamic-pituitary-gonadal axis, enabling a more robust testosterone response to acute resistance exercise (35).

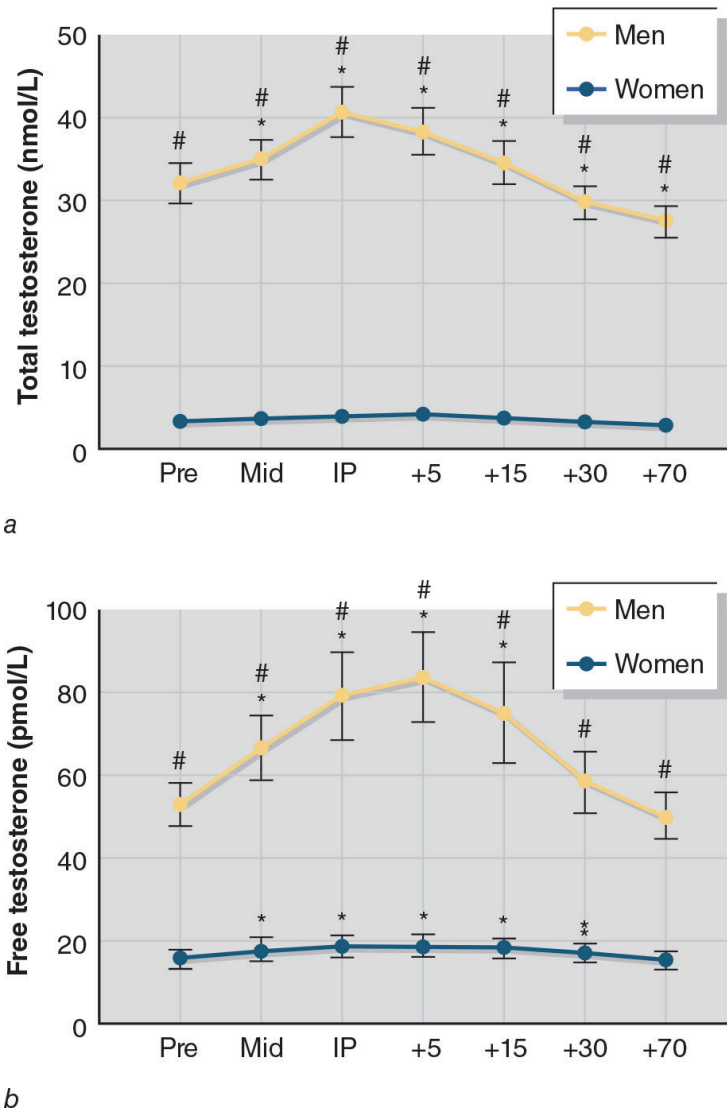


FIGURE 4.5 Responses of (a) total testosterone and (b) free testosterone in response to six sets of squats at 80% of the 1RM with 2 minutes rest between sets. The midpoint (Mid) sample is after three sets. * indicates a significant increase from pre-exercise values. # indicates a significant difference from women at the corresponding time point.

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Training Adaptations of Testosterone and Impact on Androgen Receptors

Research continues to expand our understanding of how testosterone responds to resistance training, revealing that testosterone levels dynamically respond to exercise demands,

influencing receptor activity and adaptations in different tissues like nerves and satellite cells (49, 88, 126, 133, 135). Traditional expectations that testosterone consistently increases in resting states may be outdated; however, exercise-induced testosterone levels are expected to rise due to enhanced functional capacity and workload during training sessions. These hormonal changes may be linked to neural adaptations that enhance performance. Häkkinen and colleagues (49) noted little change in muscle fiber size despite the increases in testosterone, follicle-stimulating hormone, and luteinizing hormone. Such data demonstrate that the improvements were more likely driven by neural factors augmented by hormonal changes rather than significant muscle growth.

Over time, factors such as training duration and experience play critical roles in hormonal adjustments. For instance, a notable study by Häkkinen and colleagues (49) observed small but significant increases in resting serum testosterone levels over two years among elite weightlifters, alongside increases in follicle-stimulating hormone and luteinizing hormone, which are crucial in regulating testosterone production.

Testosterone also supports the development of the nervous system and enhances neural adaptations necessary for strength gains, particularly in highly trained athletes (51, 52). Additionally, training sessions that acutely elevate testosterone levels have been linked to more remarkable resistance training adaptations (53, 119). The effects of resistance training on androgen receptors—testosterone's binding sites—are emerging as an area of significant interest, with studies showing both increases and decreases in androgen receptor content and DNA binding, depending on the exercise protocol and timing of tissue sampling (12, 93, 118, 132, 133, 136). For example, Cardaci and colleagues noted that androgen receptor-DNA binding increased only after high-intensity total body resistance exercises and not after low-intensity exercises (12). Despite these variations, resistance training generally appears to boost muscle androgen receptor content, thereby enhancing

testosterone use. The training effects for women remain unclear because only small increases in the exercise-induced response have been noted. Yet the androgen receptors in trained women rapidly respond to the available testosterone after exercise (111, 133). Nutritional strategies, particularly consuming proteins and carbohydrates before a workout, have been shown to upregulate skeletal muscle androgen receptor content, emphasizing the importance of preworkout nutrition (93). While most exercise studies have evaluated fasted states, interestingly, as pre- and post-protein and carbohydrate supplementation are used, a reduction in the testosterone values is observed, most likely due to the greater uptake by the muscle for use.

Growth Hormone

Growth hormone (GH) is produced in the pituitary gland by somatotroph cells, divided into two types based on their secretion patterns and the forms of GH. Band 1 somatotrophs generate the smaller, newly synthesized 22 kD molecular weight isoform of GH, a 191 amino acid polypeptide. Conversely, band 2 somatotrophs produce aggregates of these smaller isoforms (e.g., two 22 kD units combining to form a 44 kD molecule), which are considered older and are potential GH storage forms (61). These aggregates can be released into the bloodstream, interacting with specific tissue receptors. For instance, aggregated GH binds to receptors in the tibial growth plate to stimulate bone growth. These different molecular weight variants target various tissues, including bone, muscle, liver, and immune cells, influencing glucose use, protein synthesis, lipolysis, collagen synthesis, and enhancing immune cell function.

The presence of different GH isoforms in the blood can only be distinguished through various assays, with most studies focusing on the smaller 22 kD isoform using immunoassays. This focus has shaped much of our current understanding of GH's role and function in the body, but the aggregated forms of GH might have greater biological potency and importance (61, 91). [Figure 4.6](#) outlines the basic feedback mechanisms regulating the production of GH isoforms, highlighting the complex control the pituitary gland exercises over GH secretion.

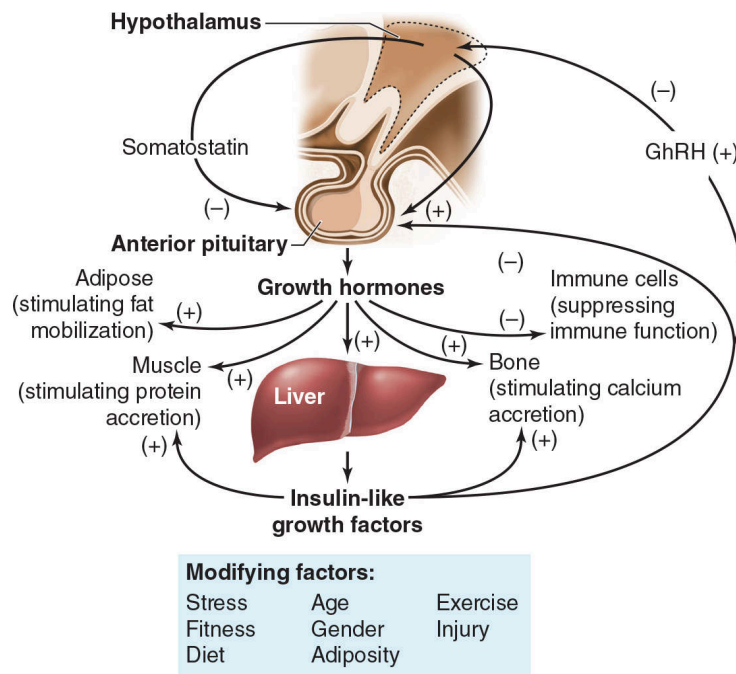


FIGURE 4.6 Diagram of the basic growth hormone cybernetics and interactions. Remember that GH will be secreted from two different types (I and II) of anterior pituitary somatotrophs, each releasing different molecular weight isoforms.

Resistance exercise increases the 22 kD isoform in both men and women. It is sensitive to the amount of total work performed and the rest period duration used because these affect the demands on the glycolytic energy system and H^+ ion production; drops in pH are a key stimulator of 22 GH release (41, 54). Women have been shown to have higher resting values of the 22 kD isoform of GH than men in the early follicular phase of the menstrual cycle (81). GH levels in women tend to be higher during the menstrual cycle's follicular phase than in the luteal phase. During the follicular phase, which occurs from the beginning of menstruation until ovulation, estrogen levels rise, and this hormonal environment is associated with increased GH secretion. While GH is present throughout the entire cycle, its secretion is generally more pronounced during this phase due to the influence of estrogen, which stimulates GH production. Additionally, oral contraceptive use can influence women's acute GH response to exercise (84). Specifically, women who used oral contraceptives exhibited a blunted or reduced GH

response compared to those not using oral contraceptives. [Figure 4.7](#) (79, 83) demonstrates the response patterns of GH to low- and high-volume exercise workouts in which exercise response patterns to different workouts are similar in men and women.

The body's response to GH varies with resistance exercise, and understanding these responses is crucial for optimizing training outcomes. Initially observed in studies from the late 1970s to early 1990s, resistance training can acutely elevate GH levels in the blood, particularly through exercises involving large muscle groups and high intensity (100, 123, 130). Research has shown that the type of workout—especially the intensity, volume, and rest periods—significantly affects acute GH responses (75, 83). Short rest periods combined with moderate resistance tend to produce higher GH spikes compared to longer rest periods.

While resting concentrations are not typically affected by training, the ability to do more total work or tolerate greater increases in H^+ ions and resultant pH decreases allows for greater peak GH concentrations in recovery. While general responses are similar to those in men, subtle differences exist due to factors such as the menstrual cycle and hormonal contraception, affecting GH levels and their response to exercise. Yet the impact on training appears minimal (79, 81, 85).

Resistance training affects the various molecular or aggregate forms of GH, with evidence showing that high-intensity workouts lead to increased production of the 22 kD GH isoform. However, research using bioassays suggests that resistance training might reduce the release of larger GH aggregates, indicating a shift toward more newly synthesized GH and less storage in aggregates (60, 72, 104). This adaptation might reflect an evolved response in the anterior pituitary gland, particularly in highly trained individuals.

Insulin-Like Growth Factors and Binding Proteins

IGFs, primarily IGF-1 and IGF-2, are small polypeptides crucial for various anabolic processes in the body, including cellular growth, development, and metabolism. These growth factors are structurally similar to insulin and originate from larger precursor molecules modified posttranslationally to become active. IGF-1, the most researched, is predominantly synthesized in the liver in response to stimulation by 22 kD GH (16, 22, 27, 91). Besides GH, IGF synthesis is also influenced by thyroid hormones and testosterone (137, 138, 139, 140).

IGFs perform several functions, including enhancing protein synthesis, increasing amino acid transport across cell membranes, and stimulating growth in tissues such as muscle and bone. In the bloodstream, IGFs are mostly bound to proteins, with only about 2% in the free (unbound) form; same as for other hormones, the binding proteins regulate IGF activity and availability. These binding proteins are critical in modulating the growth-promoting actions of the IGFs (91). Additionally, IGFs stimulate the secretion of their binding proteins within muscle cells, influencing cellular sensitivity to IGFs (105). IGF binding proteins (IGFBPs) are a group of proteins that regulate the bioavailability and activity of IGFs. There are six main types of IGFBPs (IGFBP-1 to IGFBP-6) found in humans, each with distinct binding affinities, tissue distributions, and biological functions. IGFBPs can inhibit or enhance the action of IGFs by doing the following:

1. Binding IGFs with high affinity, which prevents their interaction with IGF receptors and modulates their half-life, transport, and tissue localization
2. Regulating IGF turnover and clearance in the circulation

3. Interacting with cellular components, influencing various cellular processes independently of IGFs

IGFBPs are involved in numerous physiological processes, including growth regulation, cell proliferation, and differentiation.

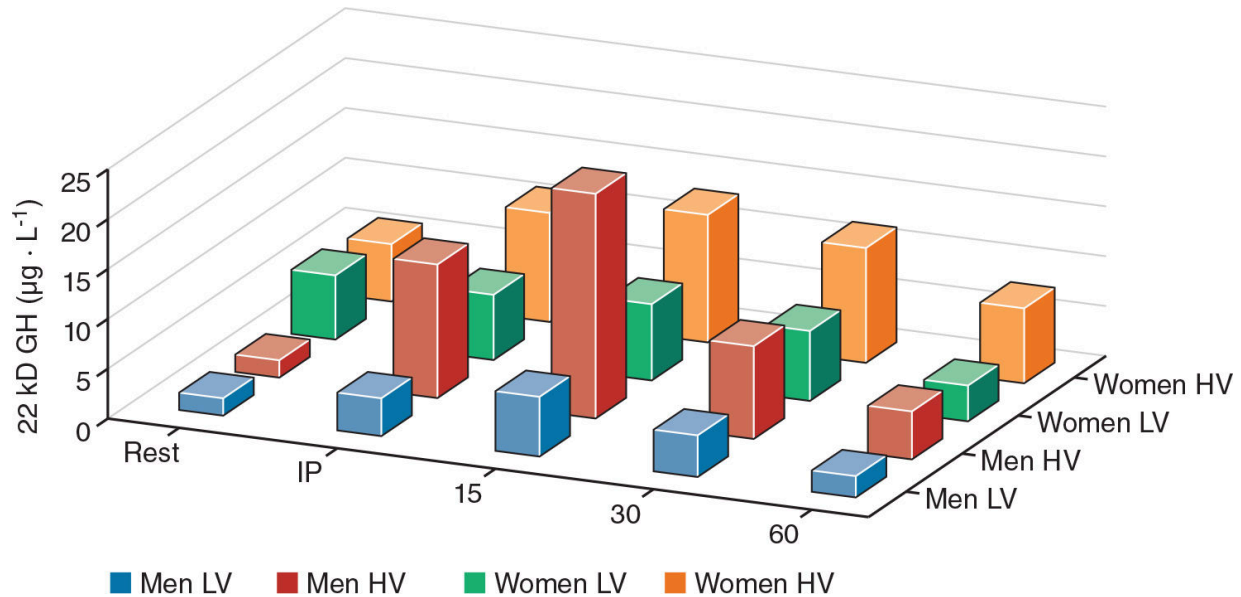


FIGURE 4.7 The 22 kD growth hormone (GH) normalized responses to high-volume (HV) and low-volume (LV) total body workout of eight resistance exercises. Men: LV (49KJ) and HV (60KJ), and women: LV (25KJ) and HV (32KJ). LV used 3-minute rest periods and a 5RM, and HV used 1-minute rest periods and a 10RM.

Data normalized from Kraemer et al. (1993) and Kraemer et al. (1990).

IGF Responses to Exercise

The role of IGFs extends to exercise response, where resistance exercise particularly influences their production and activity. Mechanical stress from resistance exercise triggers muscle fibers to produce IGF-1 locally, which then acts to stimulate muscle growth and repair. This local production is often more critical than the circulating levels of IGF-1, which may not consistently change with resistance exercise (109, 113). The response of IGF-1 to heavy resistance exercise can vary, influenced by the initial levels of IGF-1 before training and the response of GH. Studies indicate that lower baseline IGF-1 levels typically lead to increases after exercise, whereas higher baseline levels may result in no change or even decreases (W.J. Kraemer, unpublished data). Nutritional status, including caloric intake and restriction, also significantly modulates resting and exercise-induced IGF-1 concentrations (55, 92). In women, resistance training has increased total IGF-1 levels and decreased IGF-binding protein-1 during workouts, suggesting exercise-specific adaptations in the IGF system (43). The IGF superfamily's complex response patterns to training involve mechanisms such as IGF type, release, transport, and receptor interactions, along with interactions with other anabolic hormones targeting similar outcomes such as protein synthesis. However, highly trained men and women both showed increases in IGF-1 concentrations, but women demonstrated values higher than men at all time points. In these trained men and women, IGFBP-1 and 2 showed increases with exercise with no sex-dependent differences, yet IGFBP-3 concentrations were higher in women at all time points with no exercise-induced changes (54).

Training Adaptations and IGF and Binding Proteins

The full scope of training-induced IGF-1 adaptations across different tissues continues to be a significant area for research (91, 112). Training adaptations from resistance exercise influence IGF levels and their interactions at the cellular level. Resistance training enhances the activation and expression of IGF-1 receptors, boosting the signaling pathways essential for cell growth and survival. However, the full scope of how different training regimens affect IGF levels, particularly the balance between local (muscle) and systemic (circulating) IGF, remains a rich area for research. As noted in a study by Hatfield and colleagues, in highly resistance-trained men and women (i.e., squat 1RM: 2.1 ± 0.4 and $1.7 \pm 0.6 \times$ body mass, respectively), training affects the resting and acute responses of IGF-1 and IGFBPs (54). Thus, IGF-1 and binding proteins have training-related adaptations in their response patterns for production, release, binding, and interactions with cellular targets (42, 54, 91, 112, 121). Continued research is needed to better understand the complex dynamics of IGFs in response to resistance training, including the specific effects of exercise on different forms of IGF and their binding proteins. This research will help optimize training strategies and potentially inform therapeutic interventions targeting muscle health and performance (102, 106).

Adrenal Gland, Cortisol, and Its Role in Resistance Exercise

The adrenal gland is integral to the body's adaptation to stress, particularly during and after exercise. Structurally, it consists of two main components: the medulla and the cortex, each responding to stress in distinct ways. The adrenal medulla, the inner core of the gland, is directly stimulated by the autonomic nervous system during acute stress. It rapidly secretes catecholamines such as epinephrine and norepinephrine, alongside dopamine and specific enkephalin-containing polypeptides like peptide F, which are implicated in immediate stress responses and immune enhancement (129). The adrenal cortex is stimulated by adrenocorticotrophic hormone (ACTH) from the anterior pituitary gland to produce and release cortisol. Cortisol has critical roles in modulating inflammation and managing metabolic responses to stress. As such, cortisol is regulated on a systemic level by the hypothalamic-pituitary-adrenal (HPA) axis and locally within tissues by enzymes such as 11β -hydroxy-steroid dehydrogenase (11β -HSD) to maintain glucocorticoid balance (25, 27, 99). Cortisol is crucial for sustaining energy during prolonged stress by facilitating glucose availability through selective glucose sparing and mobilization, gluconeogenesis, and fat and protein catabolism. However, prolonged elevation of cortisol levels—if not returned to baseline within 24 hours—can adversely affect the body, because cortisol in this context is considered catabolic and anti-anabolic. Sustained high cortisol can impair immune function, affect fat metabolism and deposition, reduce protein synthesis, increase protein degradation, and interfere with anabolic signaling pathways, thereby promoting a catabolic state that can undermine muscle repair and growth (75, 87, 89, 92, 129). Interference in anabolic signaling is partly due to the cortisol significantly contributing to immune suppression observed after intense exercise stress. Cortisol is crucial in “inactivating” immune cell functions during recovery

phases, vital for post-exercise muscle remodeling and overall recovery (30).

Cortisol is critical in energy balance, immune regulation, and vascular integrity within skeletal muscle. It aids in increasing energy availability and modulating immune activity during exercise. After exercise, cortisol helps reduce inflammation and muscle damage, but sensitivity to cortisol decreases within 24 hours to protect against prolonged exposure. Athletes show a refined capability to convert cortisol to cortisone, protecting tissues during exercise stress. However, overtraining can impair this conversion process, potentially inhibiting anabolic activities like muscle growth by reducing IGF-I synthesis (27, 30, 91).

Despite its reputation as a catabolic hormone, cortisol is essential for carbohydrate metabolism, particularly when muscle glycogen stores are low (25, 27, 99). It plays a role in converting amino acids to carbohydrates, increasing **proteolytic enzymes**, inhibiting protein synthesis, and suppressing glycogenesis and immune function (30, 91). Cortisol levels increase in response to resistance training, particularly with high intensity or volume involving large muscle masses. The acute hormonal response prepares the body for future physical demands and aids recovery.

The resistance exercise cortisol response differs some between men and women. Men generally experience higher increases in cortisol from certain types of training protocols, such as high-volume workouts, whereas women rapidly adjust their receptor sensitivity to hormones after exercise. Such data indicate a sex-specific adaptation to training stress (133). Emerging research shows that resistance exercise can affect glucocorticoid receptor concentrations differently across sexes, indicating potential sex differences in hormonal response to training. These findings underscore the complex interaction between cortisol and its receptors in regulating muscle adaptation and recovery (31).

Resistance training influences cortisol responses and its conversion to cortisone. While chronic high cortisol levels are

detrimental, acute increases assist in muscle remodeling by helping clear damaged proteins. Adaptations to resistance training also include enhanced cortisol conversion efficiency, reflecting a physiological adaptation to regular, intense exercise (75, 127).

Resistance exercise protocols using high volume, large muscle groups, and short rest periods increase serum cortisol values (88). Though chronic high concentrations of cortisol may have adverse catabolic effects, acute increases still contribute to remodeling muscle tissue and maintaining blood glucose.

Catecholamines and Their Role in Resistance Exercise and Training

Catecholamines, primarily epinephrine but also including norepinephrine and dopamine, are secreted by the adrenal medulla and play a crucial role in enhancing strength and power during resistance training. These hormones facilitate the fight-or-flight response by acting as central motor stimulators and peripheral vascular dilators, enhancing enzyme activity and calcium release in muscles (89). Notably, higher catecholamine release before and during intense resistance sessions has been linked to sustained force output, emphasizing their importance in maintaining performance (32).

Physiological Functions of Catecholamines

Catecholamines enhance resistance training outcomes by doing the following:

- Increasing force production through central mechanisms and boosting metabolic enzyme activity
- Accelerating muscle contraction rates
- Raising blood pressure
- Enhancing energy availability and muscle blood flow through vasodilation
- Augmenting the secretion rates of other anabolic hormones such as testosterone

Acute and Training-Induced Responses

The acute response to high-intensity, short-rest resistance exercises typical in bodybuilding—characterized by increased levels of norepinephrine, epinephrine, and dopamine during the session and brief recovery periods—illustrates catecholamines' role in meeting the physical demands of such protocols (86). Epinephrine levels also correlate with lactate concentrations, reflecting its involvement in

metabolic stress responses. Importantly, adrenal responses to stress are critical until the stressor is removed, with endogenous opioid peptides like proenkephalins from the adrenal medulla playing roles in immune system modulation crucial for recovery (129). Continuous high stress without varied training can lead to delayed recovery and, eventually, adrenal exhaustion, reducing the adrenal medulla's ability to secrete catecholamines effectively.

Training Adaptations and Long-Term Impacts

Heavy resistance training enhances an athlete's capacity to secrete more epinephrine during maximal efforts (87). Consistent training may decrease epinephrine responses to single sessions, such as a bench press workout (44). This adaptation suggests that regular training can modulate catecholamine responses, reducing their surge during familiar tasks to maintain a more balanced hormonal environment conducive to ongoing strength gains and recovery. Understanding the role of catecholamines in resistance training helps in designing programs that optimize hormonal responses to enhance strength, hypertrophy, and overall athletic performance. Catecholamines' ability to interact with other hormones underscores their pivotal position in the endocrine response to resistance exercise, highlighting the importance of managing exercise intensity, volume, and recovery to maximize beneficial hormonal interactions.

Training protocols must be varied to allow the adrenal gland to engage in recovery processes and to prevent the secondary responses of cortisol, which can have negative effects on the immune system and protein structures.

How Can Athletes Manipulate the Endocrine System With Resistance Exercise?

General Concepts

- The more muscle fibers recruited for an exercise, the greater the extent of the potential remodeling process in the whole muscle.
- Only muscle fibers activated by resistance training are subject to adaptation, including hormonal adaptations to stress.

To Increase Serum Testosterone Concentrations

Serum testosterone concentrations have been shown to increase acutely by using these methods independently or in various combinations:

- Large muscle group exercises (e.g., deadlift, power clean, squat)
- Heavy resistance (85% to 95% 1RM)
- Moderate to high volume of exercise, achieved with multiple sets or multiple exercises
- Short rest intervals (30-60 s)

To Increase 22 kD Growth Hormone Concentrations

Growth hormone levels have been shown to increase acutely with use of either of these methods or both in combination:

- Use workouts with higher lactate concentrations and associated acid–base disruptions; that is, use high intensity (10RM, or heavy resistance) with three sets of each exercise (high total work) and short (1 min) rest periods.
- Supplement diet with carbohydrates and protein before and after workouts.

To Optimize Responses of Adrenal Hormones

- Use high volume, large muscle groups, and short rest periods to expose the body to adrenergic stress.
- Be careful to vary the training protocol and the rest period length from short to long over time.
- Provide days of complete rest, and use lower-volume workouts to allow the adrenal gland to engage in recovery processes to reduce stress on the adrenal medulla so as not to experience adrenergic exhaustion and to reduce stress on the adrenal cortex and prevent chronic cortisol secretions from the

adrenal cortex. This way, the stress of the exercises will not result in a nonfunctional overreaching or overtraining.

Other Hormonal Considerations in Resistance Training

A variety of hormones contribute to normal bodily functions and adaptively respond to resistance training, including insulin, thyroid hormones, and beta-endorphin. These hormones play roles in growth, repair, pain relief, and managing exercise stress, though research on their specific adaptations to resistance training is limited (15, 26, 27, 28, 51, 64, 71, 75, 82, 98). For healthy individuals, significant chronic changes in the resting concentrations of insulin and thyroid hormones are generally not observed due to tight homeostatic control. However, resistance training has been noted to improve insulin sensitivity, possibly reflecting only the acute effects of recent exercise sessions (10).

Thyroid Hormones

Thyroid hormones, particularly triiodothyronine (T3) and thyroxine (T4), play a significant role in both acute and chronic responses to resistance exercise and training. Immediately following resistance exercise, T3 levels generally remain unchanged, while T4 levels might spike briefly and then decrease during subsequent rest periods (65, 107). This acute response suggests a temporary adjustment to meet the immediate metabolic demands of exercise.

In the context of long-term resistance training, transient reductions in T3 and T4 levels can occur, which typically normalize after several months (3, 116). These fluctuations indicate an initial adaptation phase in which the thyroid axis adjusts to the increased demands of regular resistance exercise.

Thyroid hormones are crucial for metabolic control and augment other hormonal responses, underscoring their importance in physiological adaptations to training. They support the energy demands of resistance training by regulating protein synthesis and mitochondrial activity, contributing to muscle repair and growth.

Specifically, T3 enhances protein synthesis by increasing the transcription of genes involved in muscle hypertrophy and repair. Additionally, T4's conversion to T3 in peripheral tissues ensures a steady supply of active hormones to meet ongoing metabolic needs.

Thyroid hormones also help regulate lipolysis and glucose metabolism, which are essential for providing the necessary energy substrates during and after resistance exercise. By modulating these processes, thyroid hormones help maintain energy balance and facilitate recovery. Furthermore, thyroid hormones interact with other endocrine responses during resistance training. For example, they can enhance the action of GH and IGF-1, both of which are vital for muscle growth and repair. This synergistic effect amplifies the overall anabolic environment, promoting more effective muscle adaptation and strength gains. Thus, thyroid hormones are integral to immediate and long-term resistance exercise adaptations. They ensure metabolic homeostasis, support energy production, enhance protein synthesis, and interact with other hormones to facilitate muscle growth and repair. Understanding their role highlights the complex hormonal interplay required for effective training adaptations and overall metabolic regulation.

Insulin

The insulin response to resistance exercise and training is crucial to metabolic regulation and muscle adaptation. Physiologically, resistance exercise induces an acute increase in insulin sensitivity, which enhances glucose uptake into skeletal muscle, facilitating glycogen synthesis and promoting anabolic processes essential for muscle repair and growth (36, 51). This response is attributed to increased GLUT4 expression on the cell surface, driven by both contraction-mediated and insulin-mediated pathways (101). Regular resistance training amplifies this effect, improving overall short- and long-term insulin sensitivity. Such data are particularly significant for metabolic health, because enhanced insulin sensitivity associated with regular resistance training can contribute to better glycemic

control and reduced risk of type 2 diabetes (96, 107). Furthermore, the increased muscle mass from long-term resistance training enhances basal metabolic rate and glucose disposal, stabilizing insulin levels and improving metabolic efficiency (115, 119). Thus, the insulin response to resistance exercise supports immediate energy needs and recovery after exercise and contributes to long-term metabolic health and disease prevention.

Conclusion

The endocrine system's interaction with the nervous, immune, and musculoskeletal systems forms a complex network crucial for optimizing health and athletic performance. Hormones coordinate a wide range of cellular activities, influencing muscle synthesis, recovery, and overall physiological well-being. The dynamic balance between anabolic hormones, which promote muscle growth, and catabolic hormones, responsible for muscle breakdown and energy mobilization is vital for responding effectively to resistance training.

Strength and conditioning professionals need a deep understanding of the endocrine system's functions to design effective training programs that maximize performance and minimize injury risks. This knowledge enables the manipulation of hormonal responses to both acute exercise and chronic training adaptations, enhancing recovery and preventing overtraining. The subtle interplay among various bodily systems and hormones has significant implications beyond athletic performance, affecting conditions such as metabolic syndrome, insulin resistance, and cardiovascular health. By leveraging insights from the endocrine system, coaches and athletes can achieve peak performance and support long-term health.

This overview highlights the sophisticated yet orderly system that mediates the body's responses to resistance exercise, emphasizing the critical role of the endocrine system in sport science and overall health management. Understanding these complex interactions is fundamental for strength and conditioning professionals and athletes aiming to optimize training outcomes and ensure sustained athletic success and well-being.

KEY TERMS

allosteric binding site
anabolic hormone

catabolic hormone
cross-reactivity
downregulation
endocrine gland
half-life
hormone
hormone–receptor complex (H-RC)
lock-and-key theory
neuroendocrine immunology
neuroendocrinology
polypeptide hormone
proteolytic enzyme
secondary messenger
steroid hormone
target tissue cell
thyroid hormone

STUDY QUESTIONS

1. In the context of resistance training, which hormone is primarily responsible for initiating muscle repair and growth by binding to specific receptors in muscle cells and promoting protein synthesis?
 - a. cortisol
 - b. insulin-like growth factor-1
 - c. epinephrine
 - d. thyroid hormone
2. Which of the following best describes the role of binding proteins in hormone activity during resistance training?
 - a. They decrease the half-life of hormones in the bloodstream, reducing their effectiveness.
 - b. They act independently of hormones to initiate muscle growth and repair.

- c. They extend the half-life of hormones in the bloodstream, enhancing their availability and effect.
 - d. They completely inhibit the interaction of hormones with their target receptors, blocking any physiological effects.
- 3. How does testosterone primarily enhance muscle growth during resistance training?
 - a. By directly binding to membrane receptors and immediately enhancing protein degradation processes.
 - b. By promoting the synthesis of catabolic hormones that break down muscle tissue.
 - c. By binding to androgen receptors and initiating signaling pathways that enhance gene expression for muscle hypertrophy.
 - d. By decreasing the synthesis and secretion of growth hormone, thus focusing the body's resources on immediate energy use.
- 4. What is the primary effect of growth hormone on muscle tissue during resistance training?
 - a. It decreases the synthesis of insulin-like growth factors, reducing muscle tissue growth.
 - b. It stimulates the production of larger growth hormone aggregates, which decrease the tissue's ability to respond to hormonal signals.
 - c. It increases the production of the 22 kD GH isoform, which is crucial for stimulating protein synthesis and muscle growth.
 - d. It acts primarily by converting muscle glycogen into glucose, thus providing immediate energy for muscle contraction.
- 5. How does cortisol function in the context of resistance training?
 - a. Cortisol primarily supports muscle growth by suppressing catabolic activity.

- b. Elevated cortisol levels from resistance training primarily enhance immediate energy availability and help in muscle remodeling.
- c. Cortisol reduces protein synthesis and muscle adaptation by promoting protein breakdown throughout resistance training.
- d. Cortisol levels are insignificant in resistance training as they do not influence muscle adaptation or recovery.